

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Jannathul Firdaus. N

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Question 1 – Backtracking Search for Doctor–Shift Assignment

Problem definition. Assign D1, D2 and D3 to Morning (M), Afternoon (A) and Night (N), with exactly one doctor per shift, such that D1 cannot work Night, D3 cannot work Morning, and D2 must work before D3.

1.1 Constraint representation

- Variables: D1, D2, D3.
- Domain for every doctor: {Morning, Afternoon, Night}.
- All-different constraint: only one doctor may occupy each shift.
- $D1 \neq \text{Night}$.
- $D3 \neq \text{Morning}$.
- D2 must be earlier than D3: Morning < Afternoon < Night.

1.2 Backtracking search with intermediate steps and constraint checks

Step	Assignment	Constraint check	Result	Action
1	D1 = Morning	D1 is not Night; shift is unused.	Valid	Continue
2	D2 = Morning	Morning already used by D1.	Invalid	Backtrack D2
3	D2 = Afternoon	Afternoon unused; D2 can be before a later D3.	Valid	Continue
4	D3 = Morning	D3 cannot work Morning.	Invalid	Backtrack D3
5	D3 = Afternoon	Afternoon already used by D2.	Invalid	Backtrack D3
6	D3 = Night	Night unused; D2=Afternoon is before D3=Night.	Valid	Solution found

1.3 Final valid schedule

Shift	Doctor
Morning	D1
Afternoon	D2

Night	D3
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Question 2 – 5×5 Grid Search

Illustrative example used below (not stated in the assessment): S=(1,1), G=(5,5), with obstacles at (2,2), (2,3), (3,2), and (4,4).

2.1 Grid representation

S	.	.	.	.
.	#	#	.	.
.	#	.	.	.
.	.	.	#	.
.	.	.	.	G

Legend: S = Start, G = Goal, # = obstacle, . = traversable cell.

2.2 Search method and evaluation function

Because every movement costs 1, BFS can determine a minimum-number-of-moves path. However, the question explicitly requests use of the Manhattan-distance heuristic and priority-queue updates. Therefore, the demonstration below uses A* search, which combines the path cost $g(n)$ with the Manhattan heuristic $h(n)$.

- $g(n)$ = cost from S to n.
- $h(n) = |\text{row}(n) - \text{row}(G)| + |\text{column}(n) - \text{column}(G)|$.
- $f(n) = g(n) + h(n)$.
- Priority queue is ordered by the smallest $f(n)$; ties are resolved by smaller $h(n)$ for a deterministic trace.
- Moves are Up, Down, Left and Right; obstacles are never inserted as traversable successors.

2.3 Step-by-step node expansion and priority-queue updates

Coordinate convention: (row, column), with rows and columns numbered 1–5.

Expansion	Expanded node	g	h	Priority queue after expansion (node:f)
1	(1,1) S	0	8	(1,2):8, (2,1):8
2	(1,2)	1	7	(2,1):8, (1,3):8
3	(1,3)	2	6	(2,1):8, (1,4):8
4	(1,4)	3	5	(2,1):8, (1,5):8, (2,4):8
5	(1,5)	4	4	(2,1):8, (2,4):8, (2,5):8

6	(2,4)	4	4	(2,1):8, (2,5):8, (3,4):8
7	(2,5)	5	3	(2,1):8, (3,4):8, (3,5):8
8	(3,4)	5	3	(2,1):8, (3,5):8, (4,3):8, (3,3):10
9	(3,5)	6	2	(2,1):8, (4,3):8, (4,5):8
10	(4,5)	7	1	(2,1):8, (4,3):8, (5,5):8
11	(5,5) G	8	0	Goal reached; stop

Optimal path for this illustrative grid: (1,1) → (1,2) → (1,3) → (1,4) → (1,5) → (2,5) → (3,5) → (4,5) → (5,5).

Path cost: 8 moves × cost 1 = 8.

Question 3 – Autonomous Rescue Robot in a Partially Observable Environment

3(a). Constraint analysis and effect on decision-making

Constraint	Effect on decision-making	Priority/handling
Four-direction movement	The action space contains only Up, Down, Left and Right. Diagonal actions are excluded.	First restrict successor generation to legal actions.
Obstacles	Blocked cells cannot be entered, so the robot must re-plan around them.	Hard constraint; reject blocked successors immediately.
Limited sensing	The robot cannot know the full environment in advance and must act using locally observed information.	Maintain a belief/knowledge state and update it after sensing.
Movement cost = 1	Each ordinary movement contributes 1 to total path cost.	Include every movement in cumulative cost $g(n)$.
Risky-zone additional cost = 2	Entering a risky cell adds 2 to the normal movement cost, making that transition cost 3 in total.	Avoid unnecessary risky cells, but compare their actual cost with longer safe alternatives.

Minimize total cost while remaining safe	The objective is not simply shortest distance; cost and safety must both influence action selection.	Use cumulative path cost and safety constraints in search.
Dynamic knowledge / online search	The map may become known only as the robot moves, so the initial plan may become invalid or suboptimal.	Re-plan whenever new information changes the reachable or cost structure.

3(b). Solution approach – Uniform Cost Search with an Online Search Agent

A suitable approach is an Online Uniform Cost Search (UCS) agent. UCS expands the currently known lowest-cost partial route. It is appropriate because movement and risk costs are non-negative and the problem explicitly requires minimizing total cost. The online component handles the fact that the robot discovers the environment incrementally.

1. Initialize the robot at S with cumulative cost $g(S)=0$ and an initial knowledge map containing S and whatever adjacent cells can be sensed.
2. Sense the adjacent cells before selecting the next action. Mark observed obstacles as blocked and observed risky cells with their additional cost.
3. Generate only legal actions: Up, Down, Left and Right. Reject cells that are known to be blocked.
4. Assign transition cost 1 for a normal cell and 3 for entering a risky cell (1 movement cost + 2 risk cost).
5. Insert newly discovered states into a priority queue ordered by cumulative path cost $g(n)$.
6. Expand the lowest-cost reachable state and continue until a route to the currently known location of G is available.
7. Execute the next action rather than assuming that the whole map is known. After movement, sense again and update the knowledge map.
8. If a newly discovered obstacle or risky area changes the best route, update the frontier and re-plan from the robot's current state.
9. Terminate when the robot reaches G. The selected route should satisfy all hard constraints and have the minimum known cumulative cost under the discovered environment.

3(c). Application of AI concepts

Intelligent agent: The rescue robot is an intelligent agent because it perceives the environment through sensors, maintains information about what it has observed, chooses actions, moves through the environment, and updates its knowledge from new percepts. Its objective is to reach the survivor safely while minimizing cost.

Rationality: A rational agent selects an action that maximizes the expected achievement of its performance objective given its available information. Here, rational behavior means selecting a safe, legal action with the lowest appropriate expected cumulative cost, while recognizing that the robot has incomplete information.

Environment type: The environment is partially observable because the robot can observe only adjacent cells. It is sequential because each movement affects the state in which later decisions are made. It is dynamic from the agent's operational perspective because the robot must update its knowledge as it discovers previously unknown obstacles/risk information. It is discrete because states and actions are represented as grid cells and four movement actions. The task is also cost-sensitive because actions have different cumulative costs.

3(d). Justification of the chosen approach

- UCS directly accounts for the problem's cost structure. A normal movement costs 1, while entering a risky zone costs 3 in total, so the algorithm compares complete accumulated costs rather than merely counting steps.
- UCS is complete when the relevant branching structure is finite and action costs are positive, and it returns a least-cost path under the supplied cost model.
- The online-search component addresses partial observability. The robot does not need a complete map before starting; it can act, sense, update its knowledge, and re-plan.
- Hard constraints such as obstacles are handled by preventing illegal transitions, while risky cells are represented through higher transition costs.
- This approach is therefore aligned with the assessment requirements: safe navigation, dynamic knowledge updating, cost minimization, and decision-making under limited information.